



HITEC UNIVERSITY TAXILA CANTT
BS CYBER SECURITY
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AI LAB REPORT:2

SUBMITTED BY:

HALIMA IMRAN

24-CYS-033

SUBMITTED TO:

SIR MUBASHIR IQBAL

LAB REPORT:2

QUESTION NO:1

Count the number of strings where the string length is 2 or more and the first and last character are same from a given list of strings. Sample List: ['abc', 'xyz', 'aba', '1221', 'xyzzyx', 'aa', '122'] Expected Result: 4 .

CODE:

```
# Sample list
words = ['abc', 'xyz', 'aba', '1221', 'xyzzyx', 'aa', '122']

count = 0

for word in words:
    if len(word) >= 2 and word[0] == word[-1]:
        count += 1

print("Number of strings with same first and last character:", count)
```

OUTPUT:

```
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/P
Number of strings with same first and last character: 4
PS C:\Users\HP>
```

QUESTION NO:2

Find the sum and product of numbers in a tuple. Sample List: (1, 2, 4, 2, 6), Sum is: 15, Product is: 96.

CODE:

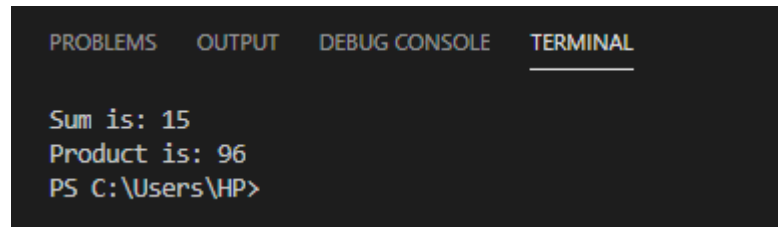
```
numbers = (1, 2, 4, 2, 6)

# Calculate sum
sum_of_numbers = sum(numbers)

# Calculate product
product_of_numbers = 1
for num in numbers:
    product_of_numbers *= num

print("Sum is:", sum_of_numbers)
print("Product is:", product_of_numbers)
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

Sum is: 15
Product is: 96
PS C:\Users\HP>
```

QUESTION NO:3

Write a program to print largest number in a List Sample List:
(1, 2, 4, 2, 6).

CODE:

```
numbers = [1, 2, 4, 2, 6]

# Find the largest number
largest = max(numbers)

# Display the result
print("The largest number in the list is:", largest)
```

OUTPUT:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe C:/Users/HP/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe
The largest number in the list is: 6
PS C:\Users\HP>

```

QUESTION NO:4

Write a Python program to remove an empty tuple(s) from a list of tuples. Sample L = [(), (), (' '), ('a', 'b'), (), ('a', 'b', 'c'), (), ('d')], Result = [(' '), ('a', 'b'), ('a', 'b', 'c'), ('d')].

CODE:

```

L = [( ), ( ), (' '), ('a', 'b'), ( ), ('a', 'b', 'c'), ( ), ('d',)]

# Remove empty tuples using list comprehension
result = [t for t in L if t]

print("Result =", result)

```

OUTPUT:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

Module PSReadLine'.

PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe C:/Users/HP/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe
Result = [(' '), ('a', 'b'), ('a', 'b', 'c'), ('d',)]
PS C:\Users\HP>

```

QUESTION NO:5

Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form ('x', x*x).
Sample: Input a number 4 Output: {'1': 1, '2': 4, '3': 9, '4': 16}.

CODE:

```
n = int(input("Input a number: "))

# Create dictionary using loop
result = {}
for x in range(1, n + 1):
    result[str(x)] = x * x

# Print the result
print(result)
```

OUTPUT:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
```

```
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python39-64/Python.exe C:/Users/HP/AppData/Local/Programs/Python/Python39-64/Scripts/python.exe
Input a number: 4
{'1': 1, '2': 4, '3': 9, '4': 16}
PS C:\Users\HP> 
```

QUESTION NO:6

Create a program using Dictionary that takes Numbers as input and print it in words form Sample: Input: 1234, Output: One Two Three Four.

CODE:

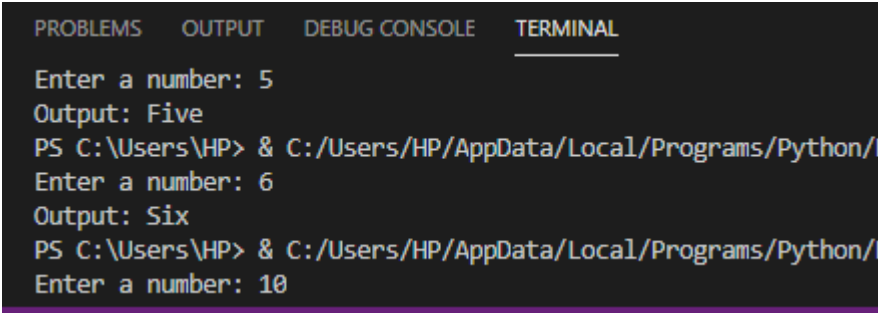
```
num_to_word = {
    '0': 'Zero',
    '1': 'One',
    '2': 'Two',
    '3': 'Three',
    '4': 'Four',
    '5': 'Five',
    '6': 'Six',
    '7': 'Seven',
    '8': 'Eight',
    '9': 'Nine'
}

# Taking number input from the user
number = input("Enter a number: ")

# Converting each digit to its word form
output = ""
for digit in number:
    if digit in num_to_word:
        output += num_to_word[digit] + " "

# Display result
print("Output:", output.strip())
```

OUTPUT:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL
Enter a number: 5
Output: Five
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/
Enter a number: 6
Output: Six
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/
Enter a number: 10
```

CONCLUSION:

In this lab, we learned how to work with **lists, tuples, and dictionaries** in Python to perform various operations. We practiced programs such as removing empty tuples, finding the largest number, calculating sum and product, counting specific strings, generating square dictionaries, and converting numbers into words.

Through these exercises, we improved our understanding of **loops, conditions, and data structures** like **lists, tuples, and dictionaries**. This lab helped us develop logical thinking and apply Python concepts to solve real-world problems effectively.
